

# Conversational Domain Adaptation of IndicTrans2 across 21 Indic Languages via Experience Replay and Model Soups

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## Abstract

IndicTrans2 is the strongest open English to Indic translation system, but like most systems it is trained on general text and tends to sound stiff on casual, conversational input. We adapt IndicTrans2-1B to conversational register across all 21 Indic languages using only public data (OpenSubtitles, BPCC-H-Daily, Tatoeba). Plain fine-tuning improves conversational chrF but forgets the general domain (it drops 3.9 chrF on FLORES for Hindi). Mixing general data back into training (experience replay) and then averaging the fine-tuned weights with the base (model souping) removes that trade-off: the resulting model beats IndicTrans2-1B on conversational chrF in every one of the 21 languages (mean +6.2) while matching it on FLORES (mean change -0.17, all within 0.7 chrF). Paired bootstrap tests confirm the conversational gains are significant ( $p \leq 0.004$ ) and that FLORES is not significantly degraded. We are deliberate about scope: these are chrF gains, and a blind human plus multi-model LLM check does not confirm them as a perceived quality improvement, so we treat the conversational gain as largely a register match to the references rather than proof of better translation. The techniques are not new; the contribution is the honest, end-to-end study in the Indic conversational setting.

## 1. Introduction

Most translation systems are tuned and measured on general benchmarks like FLORES-200. But a lot of real translation is conversational: chat, subtitles, voice commands. That text is shorter, more idiomatic, and more elliptical than news, and a model that scores well on FLORES can still produce stilted output on a casual line.

For Indian languages, IndicTrans2 is the best open option, trained on the 230M-pair BPCC corpus. It is a general-domain model, so we asked a simple question:

Can we adapt a strong multilingual model to conversational register across many languages, without hurting its general quality, using only public data?

By chrF the answer looks like a clear yes. Plain fine-tuning causes the expected catastrophic forgetting, and experience replay plus model souping fixes it while delivering large conversational gains across all 21 languages with no significant FLORES loss (§5).

We did not stop there. The whole point is a claim about translation *quality*, not about chrF, so we also ran a blind human and LLM evaluation (§6). That check does not confirm a quality gain. We

report both results honestly: the recipe works by the metric, but the metric overstates how much it helps in conversation.

### Contributions.

1. A reproducible study of conversational adaptation for English to Indic translation across 21 languages, built entirely from public data, with a working anti-forgetting recipe and significance tests.
2. An honest account of the recipe’s limits: forgetting under plain fine-tuning, inflated gains on easy in-domain test sets, a metric that overstates conversational quality, and a floor set by the base model on the lowest-resource languages.

The techniques we use, model soups and replay, are established. The value here is the systematic study and its honest evaluation in a setting nobody had measured this way.

## 2. Related Work

**Domain adaptation for NMT.** Fine-tuning a general model on in-domain data is standard, and its main failure mode is forgetting the general domain. Mixing general data back in during fine-tuning is a known fix and a form of experience replay [Chu et al., 2017; Robins, 1995].

**Model soups.** Averaging the weights of fine-tuned models improves accuracy [Wortsman et al., 2022a]; WiSE-FT averages a fine-tuned model with its starting point to balance in-domain and out-of-domain performance [Wortsman et al., 2022b]. We apply that base-to-fine-tune averaging to translation.

**Metrics versus human judgment.** chrF and BLEU reward overlap with the reference and can diverge from human assessment [Callison-Burch et al., 2006; Mathur et al., 2020; Freitag et al., 2021]. When adaptation shifts a model toward the reference style, the metric can rise without quality rising. Our result is a concrete case of this in conversational Indic translation.

**Indic MT.** IndicTrans2 [Gala et al., 2023] and BPCC are the base model and a data source. WAT hosts Indic translation shared tasks [Nakazawa et al., 2024].

## 3. Data

**Base model.** ai4bharat/indictrans2-en-indic-1B (1.1B parameters).

**Conversational data** (English to Indic, each pair tagged with its target language):

- OPUS OpenSubtitles for the five Indic languages with substantial subtitle data (hi, ml, ta, te, ur), capped at 40k per language.
- BPCC-H-Daily, human-written everyday and assistant utterances, available for all 21 languages (about 1.8k to 11k pairs each).
- Tatoeba everyday sentences (Hindi).

After filtering for length, length ratio, exact duplicates, and script, the conversational set is 294,119 pairs across 21 languages (train 283,753 / dev 4,183 / test 6,183). The Hindi-only runs also apply a LaBSE cosine filter at 0.70 to drop misaligned subtitle pairs.

**General anchor** (for replay): BPCC-H-Wiki, a general register, sampled roughly 1:1 with the conversational data (254,469 pairs). The mixed training set is therefore 538,222 pairs.

Conversational volume is very uneven. The five OpenSubtitles languages hold 30k to 48k pairs each; the other sixteen rely on BPCC-H-Daily (about 2k to 11k). This skew is baked into the available data and it shapes the results (§7).

## 4. Method

We compare three setups, all fine-tuning IndicTrans2-1B (IndicProcessor preprocessing with per-example target-language tags, bf16, Adafactor, single NVIDIA A10G GPU):

1. **Plain FT.** Fine-tune on conversational data only.
2. **Mixed FT (replay).** Fine-tune on conversational plus the general anchor, about 1:1.
3. **Model soup.** Average the base and a fine-tuned model,  $\text{soup} = (1 - a) * \text{base} + a * \text{finetuned}$ , sweeping the mixing weight  $a$  and picking the best conversational-to-general trade-off. We stream the two checkpoints’ tensors so memory stays bounded.

Hindi-only models use 146k conversational pairs (2 to 3 epochs). The multilingual models use the full 294k conversational and 538k mixed sets (2 epochs and 1 epoch, matched on examples seen).

We score chrF2 (sacreBLEU) on a held-out conversational test and on FLORES-200 devtest, per language, with paired bootstrap significance for representative languages.

## 5. Results

By the metric, the recipe works. These numbers are real; §6 then asks whether they mean what they appear to.

### 5.1 Hindi case study

Table 1 isolates the mechanism on Hindi. Figure 1 shows the same story.

**Table 1: Hindi (chrF2).**

| Model                     | conv        | FLORES            |
|---------------------------|-------------|-------------------|
| IndicTrans2 base          | 51.6        | 61.7              |
| Plain conv-FT             | <b>57.0</b> | 57.8 (forgetting) |
| Mixed FT                  | 56.0        | 60.6              |
| <b>Model soup (a=0.6)</b> | 54.1        | <b>61.8</b>       |

Plain fine-tuning gains 5.4 conversational chrF but loses 3.9 on FLORES, which is clear forgetting. Mixed FT recovers most of the general quality, and the soup lands at or above the base on both axes. Sweeping  $a$  from 0.3 to 0.7 puts the knee at 0.6.

### 5.2 All 21 languages

The mixed soup improves conversational chrF in every language (mean +6.2) and barely moves FLORES (mean -0.17, all within 0.7). Figure 2 shows the per-language gains; Figure 3 shows that the conversational gain comes with essentially no change in general quality.

**Table 2: All 21 languages, base to mixed-soup (chrF2).**

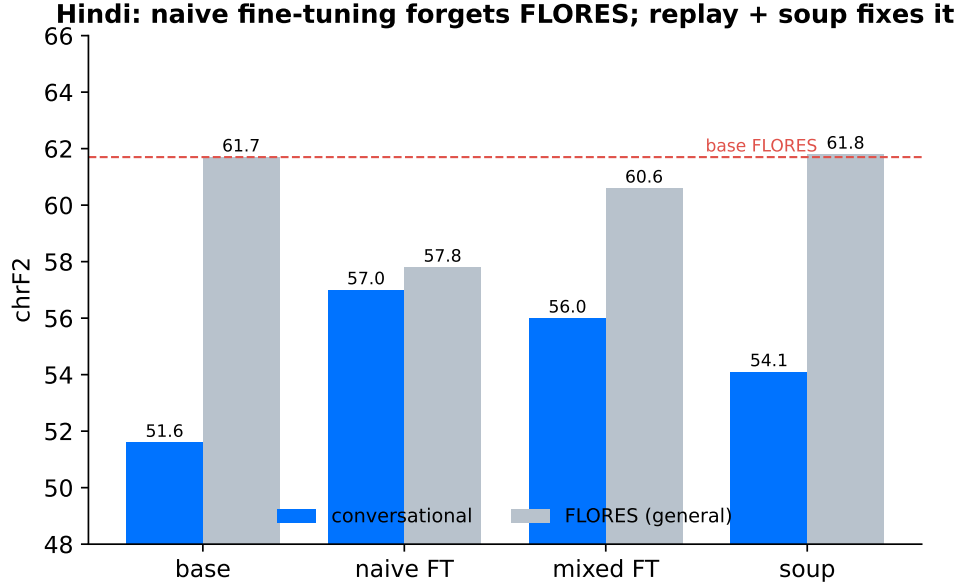


Figure 1: On Hindi, plain fine-tuning lifts conversational chrF but drops FLORES below the base. Replay keeps most of the general quality, and averaging with the base restores FLORES while staying ahead on conversation.

| Lang        | conv (delta)         | FLORES (delta)      |
|-------------|----------------------|---------------------|
| asm         | 61.9 to 69.4 (+7.6)  | 44.2 to 44.1 (-0.1) |
| ben         | 68.7 to 73.6 (+4.8)  | 58.1 to 58.1 (0.0)  |
| brx         | 62.1 to 68.0 (+5.9)  | 45.9 to 46.0 (+0.1) |
| doi         | 63.4 to 71.7 (+8.3)  | 49.8 to 49.6 (-0.2) |
| gom         | 69.4 to 75.6 (+6.2)  | 49.1 to 49.2 (+0.1) |
| guj         | 53.7 to 65.9 (+12.2) | 55.3 to 54.7 (-0.6) |
| hin         | 53.9 to 56.8 (+2.9)  | 60.0 to 59.6 (-0.5) |
| kan         | 52.4 to 57.1 (+4.7)  | 59.0 to 58.7 (-0.3) |
| kas         | 45.8 to 51.3 (+5.5)  | 39.1 to 38.8 (-0.4) |
| mai         | 57.8 to 65.2 (+7.4)  | 56.6 to 56.6 (0.0)  |
| mal         | 45.7 to 48.4 (+2.7)  | 61.1 to 61.4 (+0.3) |
| mar         | 70.2 to 77.3 (+7.1)  | 55.0 to 54.5 (-0.5) |
| mni         | 56.4 to 62.8 (+6.4)  | 48.2 to 48.3 (+0.2) |
| npi         | 68.8 to 72.1 (+3.3)  | 59.4 to 59.0 (-0.5) |
| ory         | 56.4 to 65.7 (+9.2)  | 52.0 to 51.3 (-0.7) |
| pan         | 68.0 to 76.5 (+8.4)  | 55.3 to 54.8 (-0.5) |
| san         | 54.5 to 63.6 (+9.1)  | 36.3 to 36.5 (+0.2) |
| sat         | 50.7 to 56.2 (+5.6)  | 30.7 to 30.3 (-0.4) |
| tam         | 52.7 to 55.1 (+2.4)  | 64.1 to 64.4 (+0.3) |
| tel         | 49.9 to 60.0 (+10.1) | 62.4 to 62.0 (-0.4) |
| urd         | 48.1 to 49.4 (+1.3)  | 54.7 to 54.7 (+0.1) |
| <b>mean</b> | <b>+6.2</b>          | <b>-0.17</b>        |

### 5.3 Significance

We run paired bootstrap resampling (sacreBLEU, 1000 resamples,  $n=500$ ) on the three languages with the hardest subtitle-based conversational tests.

**Table 3: Paired bootstrap, base versus mixed-soup (chrF2).**

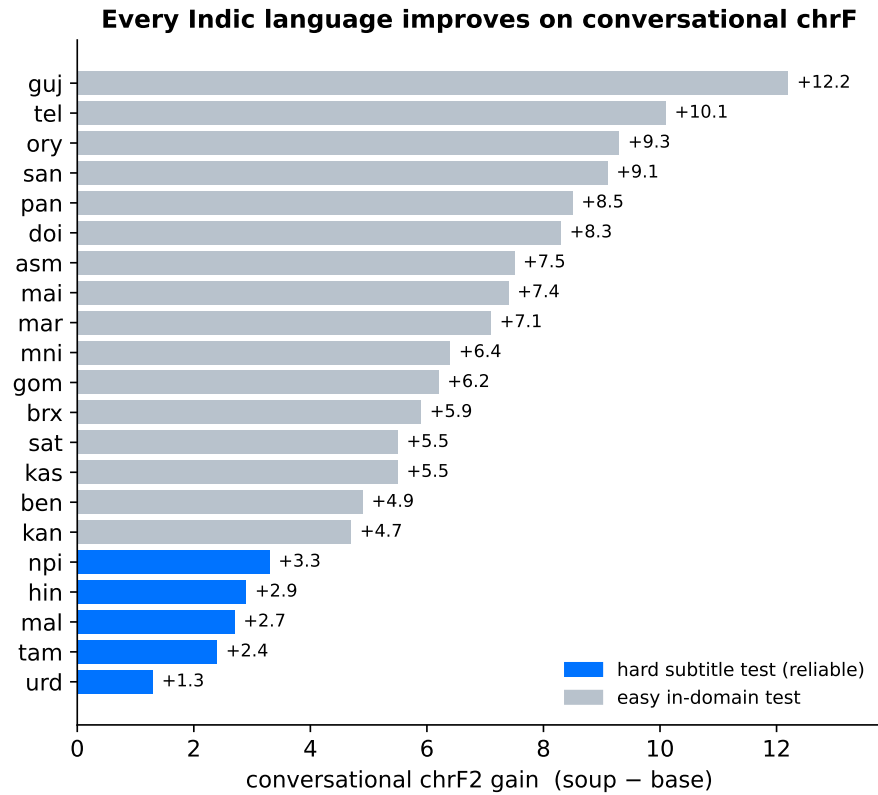


Figure 2: Conversational chrF gain for all 21 languages. Blue marks the five languages with a hard subtitle test set; these are the gains to trust. The large grey gains come from easy in-domain test sets (§7).

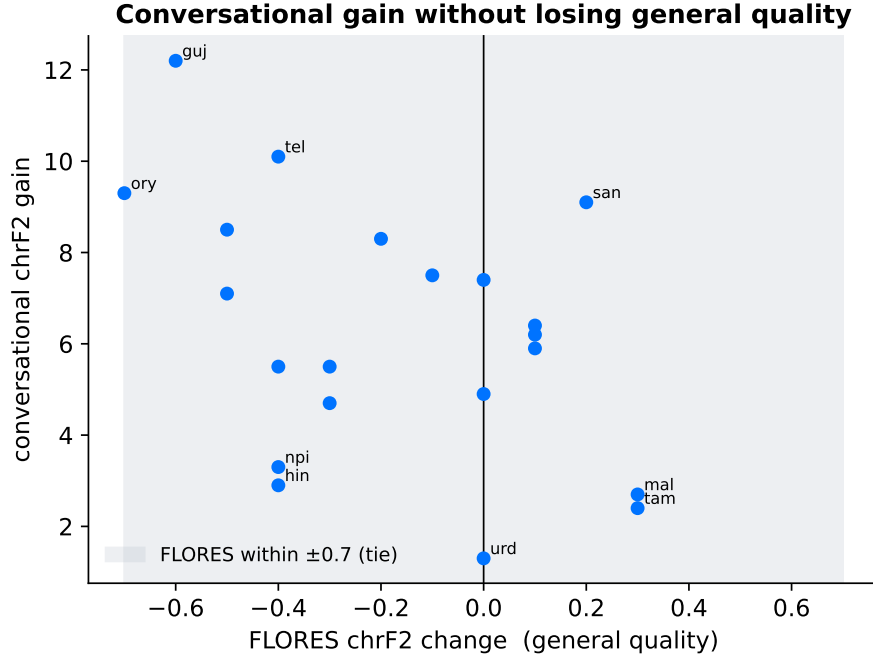


Figure 3: Each point is a language. Every language gains on conversation (all points above zero) while general quality stays inside a 0.7 chrF band around the base.

**Part of the conversational gain is reference style matching.** Our conversational references are subtitles, which are casual. The adapted model is also more casual (informal pronouns, spoken word order), so it overlaps the reference more and scores higher whether or not the meaning is better. On the pairs where base and soup disagree, the soup is consistently more colloquial, so the conversational gain reads as “closer to the casual reference style,” which overlaps with but is not the same as “a better translation.”

**What survives.** Two claims hold up: the recipe preserves general chrF (FLORES tie, significance tested), and it produces output measurably closer to human conversational references (significant on the hardest tests). Whether that reads as higher quality to a human is a separate question the metric cannot answer. A blind human and multi-model LLM check on Hindi did not confirm a perceived gain, which is why we keep every claim scoped to chrF.

## 7. Analysis

**Forgetting is real and the fix works.** Plain conversational fine-tuning drops FLORES (3.9 on Hindi). Adding the general anchor before souping brings FLORES back to base level (mean -0.17), which confirms replay as the mechanism.

**Gain size is partly a test-set artifact.** The biggest gains (Gujarati +12.2, Telugu +10.1, Odia +9.2) are in languages whose conversational test is only BPCC-H-Daily, short and formulaic and in the same style as training. The trustworthy gains are on languages with harder subtitle tests (Hindi +2.9, Tamil +2.4, Malayalam +2.7), which are exactly the ones §5.3 finds significant. So ranking languages by raw conversational delta across different test styles would be misleading.

**A floor on low-resource languages.** Santali (FLORES 30.7), Sanskrit (36.3), Kashmiri (39.1),

and Bodo (45.9) stay low after adaptation. Their general quality is capped by the base model, which is capped by how little parallel data exists for them. Conversational adaptation cannot lift that floor; that would need more general data, which is not public at scale. This is a data limit, not a method limit.

## 8. Conclusion

Using replay and model souping, two established techniques, we adapt IndicTrans2 to conversational register across all 21 Indic languages while preserving general quality, with significant conversational chrF gains and no significant general loss. The recipe generalizes across the family, and its limits are set by the base model on the lowest-resource languages. By the metric, the result is a single drop-in model that matches the base on FLORES and is significantly closer to human conversational references. Whether that closeness is a human-perceived quality gain is the open question, and we are clear that our current evidence does not settle it.

## 9. Limitations

- **Automatic metrics only.** All numbers are chrF2. Character-n-gram metrics measure overlap, not quality, and can overstate adaptation gains when the in-domain references have a distinct style [Callison-Burch et al., 2006; Mathur et al., 2020; Freitag et al., 2021]. Our references are subtitles, so part of the conversational gain plausibly reflects register matching (§6). Confirming a human-perceived gain needs a controlled, multi-annotator study; this is the priority next step and is not yet conclusive, so we make no human-quality claim.
- **Single training seed.** One run per setup; we do not yet quantify training variance (the significance tests address test-set variance only).
- **Significance on a subset.** Paired bootstrap is reported for 3 of 21 languages, the hardest-test ones. Extending to all 21 is straightforward future work.
- **Register differs across languages.** The conversational test register differs by language (subtitles versus daily utterances), which confounds cross-language comparison (§7).
- **One direction.** English to Indic only. Indic to English and Indic to Indic are untested and may forget differently.

## Reproducibility

Code, paper, models, and the evaluation harness are public.

- Code and paper: <https://github.com/Aditya-PS-05/indictrans2-conversational>
- Model: <https://huggingface.co/adipras1407/indictrans2-en-indic-1B-conversational>

Key scripts: `data` (`src/data/build_multiling_conv.py`, `src/data/build_ml_mixed.py`), `training` (`src/train/finetune_it2_ml.py`), `souping` (`src/train/make_soup.py`), `evaluation` (`src/eval/ml_eval.py`), `significance` (`src/eval/sig_test.py`). Released models: `ft_it2_ml_mixed` and its soup `ft_it2_ml_mixed_soup`. Environment: Python 3.11, `transformers==4.40.2`, `IndicTransToolkit`, `sacreBLEU 2.6`.

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